

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

CSA5112 – CRYPTOGRAPHY AND NETWORK SECURITY

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MCQ

1. Which property ensures the same input always produces the same hash?

- ☐ A) Avalanche effect
- ☒ B) Determinism
- ☐ C) Collision resistance
- ☐ D) Diffusion

Correct Answer: B) Determinism

2. What is the primary purpose of SHA-3?

- ☐ A) Faster encryption
- ☒ B) Alternative secure hashing standard
- ☐ C) Data compression
- ☐ D) Key exchange

Correct Answer: B) Alternative secure hashing standard

3. Which attack tries all possible inputs to match a hash?

- ☐ A) Replay attack
- ☒ B) Brute-force attack

☐ C) MITM attack

☐ D) DoS attack

Correct Answer: B) Brute-force attack

4. What is a salted hash?

☐ A) Encrypted hash

☒ B) Hash with random data added

☐ C) Shortened hash

☐ D) Reversed hash

Correct Answer: B) Hash with random data added

5. Which algorithm is widely used for password hashing?

☐ A) AES

☐ B) RSA

☒ C) bcrypt

☐ D) DES

Correct Answer: C) bcrypt

6. What does second pre-image resistance ensure?

☐ A) Same input produces same output

☒ B) Hard to find different input with same hash

☐ C) Fast hashing

☐ D) Small hash size

Correct Answer: B) Hard to find different input with same hash

7. What is the output size of SHA-512?

- ☐ A) 256 bits
- ☐ B) 128 bits
- ☒ C) 512 bits
- ☐ D) 1024 bits

Correct Answer: C) 512 bits

8. Which technique slows down brute-force attacks?

- ☐ A) Encryption
- ☐ B) Key exchange
- ☒ C) Key stretching
- ☐ D) Compression

Correct Answer: C) Key stretching

9. What is the purpose of a nonce?

- ☐ A) Encryption
- ☒ B) Prevent replay attacks
- ☐ C) Compress data
- ☐ D) Generate keys

Correct Answer: B) Prevent replay attacks

10. Which function is irreversible?

- ☐ A) Encryption
- ☒ B) Hashing
- ☐ C) Compression
- ☐ D) Encoding

Correct Answer: B) Hashing

11. Which algorithm is used in DSS?

☐ A) AES

☒ B) DSA

☐ C) DES

☐ D) RC4

Correct Answer: B) DSA

12. What is required to verify a digital signature?

☐ A) Private key

☒ B) Public key

☐ C) Session key

☐ D) Shared key

Correct Answer: B) Public key

13. What does a digital signature guarantee?

☐ A) Confidentiality

☒ B) Integrity and authenticity

☐ C) Compression

☐ D) Speed

Correct Answer: B) Integrity and authenticity

14. What happens if message digest changes?

☐ A) Signature remains valid

☒ B) Verification fails

☐ C) Encryption fails

☐ D) Speed increases

Correct Answer: B) Verification fails

15. Which attack involves forging a signature?

☐ A) Replay

☒ B) Forgery

☐ C) DoS

☐ D) Sniffing

Correct Answer: B) Forgery

16. What is the role of a hash in signatures?

☐ A) Encrypt data

☐ B) Reduce size

☒ C) Ensure integrity

☐ D) Store keys

Correct Answer: C) Ensure integrity

17. Which key is kept secret in signatures?

☐ A) Public key

☒ B) Private key

☐ C) Session key

☐ D) Shared key

Correct Answer: B) Private key

18. What ensures trust in certificates?

- ☐ A) Encryption
- ☒ B) Certificate Authority
- ☐ C) Compression
- ☐ D) Hash size

Correct Answer: B) Certificate Authority

19. What is certificate revocation?

- ☐ A) Key generation
- ☒ B) Canceling validity
- ☐ C) Encryption
- ☐ D) Compression

Correct Answer: B) Canceling validity

20. What is OCSP used for?

- ☐ A) Encryption
- ☒ B) Certificate validation
- ☐ C) Compression
- ☐ D) Routing

Correct Answer: B) Certificate validation

21. What is the main goal of Kerberos tickets?

- ☐ A) Encryption
- ☒ B) Authentication
- ☐ C) Compression

☐ D) Routing

Correct Answer: B) Authentication

22. Which protocol uses tickets for authentication?

☐ A) SSL

☒ B) Kerberos

☐ C) FTP

☐ D) HTTP

Correct Answer: B) Kerberos

23. What is a session key?

☐ A) Permanent key

☒ B) Temporary key

☐ C) Public key

☐ D) Private key

Correct Answer: B) Temporary key

24. What is mutual authentication?

☐ A) One-way authentication

☒ B) Both parties verify each other

☐ C) Anonymous login

☐ D) Weak authentication

Correct Answer: B) Both parties verify each other

25. What ensures freshness in Kerberos?

☐ A) Encryption

- ☒ B) Timestamp
- ☐ C) Compression
- ☐ D) Hashing

Correct Answer: B) Timestamp

26. What is a replay attack?

- ☐ A) Data modification
- ☒ B) Resending valid data
- ☐ C) Encryption
- ☐ D) Compression

Correct Answer: B) Resending valid data

27. What is the function of an authenticator in Kerberos?

- ☐ A) Data storage
- ☒ B) Identity proof
- ☐ C) Encryption
- ☐ D) Compression

Correct Answer: B) Identity proof

28. What happens if system clocks differ in Kerberos?

- ☐ A) Faster authentication
- ☒ B) Failure
- ☐ C) Better security
- ☐ D) No effect

Correct Answer: B) Failure

29. What is the KDC composed of?

- ☐ A) Client & Server
- ☒ B) AS & TGS
- ☐ C) Firewall & IDS
- ☐ D) Router & Switch

Correct Answer: B) AS & TGS

30. What is the AS in Kerberos?

- ☐ A) Application Server
- ☒ B) Authentication Server
- ☐ C) Access Server
- ☐ D) Admin Server

Correct Answer: B) Authentication Server

31. What is X.509 certificate format used in?

- ☐ A) Networking
- ☒ B) Digital certificates
- ☐ C) Compression
- ☐ D) Routing

Correct Answer: B) Digital certificates

32. What identifies the certificate owner?

- ☐ A) Public key
- ☒ B) Subject field
- ☐ C) Signature

☐ D) Hash

Correct Answer: B) Subject field

33. What ensures certificate integrity?

☐ A) Encryption

☒ B) Digital signature

☐ C) Compression

☐ D) Routing

Correct Answer: B) Digital signature

34. What is a root certificate?

☐ A) Expired certificate

☒ B) Self-signed trusted certificate

☐ C) Invalid certificate

☐ D) Temporary certificate

Correct Answer: B) Self-signed trusted certificate

35. What is certificate pinning?

☐ A) Storing keys

☒ B) Binding certificate to host

☐ C) Encryption

☐ D) Compression

Correct Answer: B) Binding certificate to host

36. What is the risk of weak CA validation?

☐ A) Faster speed

☒ B) MITM attacks

☐ C) Compression

☐ D) Routing

Correct Answer: B) MITM attacks

37. What is the purpose of trust chain?

☐ A) Encryption

☒ B) Validate certificate hierarchy

☐ C) Compress data

☐ D) Routing

Correct Answer: B) Validate certificate hierarchy

38. What is digital certificate used for?

☐ A) Compression

☒ B) Authentication

☐ C) Routing

☐ D) Storage

Correct Answer: B) Authentication

39. What ensures non-repudiation?

☐ A) Encryption

☒ B) Digital signature

☐ C) Compression

☐ D) Routing

Correct Answer: B) Digital signature

40. What is a hash collision?

- ☐ A) Same input
- ☒ B) Same output for different inputs
- ☐ C) Encryption
- ☐ D) Compression

Correct Answer: B) Same output for different inputs

41. Which hash is strongest?

- ☐ A) MD5
- ☐ B) SHA-1
- ☒ C) SHA-256
- ☐ D) CRC

Correct Answer: C) SHA-256

42. What is a one-way function?

- ☐ A) Reversible
- ☒ B) Irreversible
- ☐ C) Fast
- ☐ D) Small

Correct Answer: B) Irreversible

43. What ensures authentication in PKI?

- ☐ A) Encryption
- ☒ B) Certificates
- ☐ C) Compression

☐ D) Routing

Correct Answer: B) Certificates

44. What is key escrow?

☐ A) Key deletion

☒ B) Key storage by third party

☐ C) Encryption

☐ D) Compression

Correct Answer: B) Key storage by third party

45. What is identity spoofing?

☐ A) Encryption

☒ B) Fake identity

☐ C) Compression

☐ D) Routing

Correct Answer: B) Fake identity

46. What is trust anchor?

☒ A) Root CA

☐ B) Client

☐ C) Server

☐ D) Router

Correct Answer: A) Root CA

47. What ensures integrity of message?

☐ A) Encryption

- ☒ B) Hash
- ☐ C) Compression
- ☐ D) Routing

Correct Answer: B) Hash

48. What is digital envelope?

- ☐ A) Hash
- ☒ B) Encrypted message + key
- ☐ C) Compression
- ☐ D) Routing

Correct Answer: B) Encrypted message + key

49. What is certificate fingerprint?

- ☐ A) Encryption
- ☒ B) Hash of certificate
- ☐ C) Compression
- ☐ D) Routing

Correct Answer: B) Hash of certificate

50. What is PKI used for?

- ☐ A) Routing
- ☒ B) Secure communication
- ☐ C) Compression
- ☐ D) Storage

Correct Answer: B) Secure communication